

The Gradient Second-Moment Hypothesis Is Unnecessary

Maximal intersection position for log-concave functions

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Status: candidate full solution, likely valid, pending human review.

1 Source question

Hoehner and Roysdon introduce maximal intersection position for a pair of integrable log-concave functions and prove a John-type isotropicity theorem [1, Theorem 11]. Their support regularity property has three clauses; the third is

$$\int_{K_g} |\nabla g(x)|^2 dx < \infty. \quad (\text{SM2})$$

Immediately after the theorem they ask whether this assumption is necessary or merely an artifact of the proof [1, Remark 13]. The question occurs on PDF page 6:

and the references therein.

Remark 13. Many log-concave functions satisfy the second moment condition in Theorem 11, including gaussians and characteristic functions of convex bodies. It is unclear to us if this assumption is necessary, or if it is simply an artifact of our proof method. We leave this question as a topic for future research.

The following corollary is immediate from (6) and (7).

We prove that (SM2) can be deleted. The other two support regularity conditions in the source theorem remain unchanged.

2 Definitions and statement

Write $K_h = \text{supp } h$. The pair f, g is in maximal intersection position when $(I, 0)$ maximizes

$$P_{f,g}(T, z) = \int_{\mathbb{R}^n} f(x)g(T^{-1}(x - z)) dx \quad ((T, z) \in \text{SL}_n(\mathbb{R}) \times \mathbb{R}^n).$$

The two retained support conditions are

$$(R1) \quad \mathcal{H}^{n-1}(\partial K_f \cap \partial K_g) = 0;$$

$$(R2) \quad \mathcal{H}^{n-1}(\partial K_f \cap \partial(TK_g + z)) = 0 \text{ for all but finitely many } (T, z) \in \text{SL}_n(\mathbb{R}) \times \mathbb{R}^n, \text{ exactly as stipulated in the source support-regularity hypothesis.}$$

We use the same regularity and trace conventions as the source paper.

Theorem 1 (Removal of the second-moment hypothesis). *Let $f, g = e^{-\varphi}$ belong to the class $\mathcal{L}^n$ of Hoehner–Roysdon and satisfy (R1)–(R2). If f and g are in maximal intersection position, then every conclusion of [1, Theorem 11] holds, without assuming (SM2). In particular, for every matrix T ,*

$$\begin{aligned} & \int_{K_f \cap \text{int } K_g} fg \langle \nabla \varphi, Tx \rangle dx + \int_{K_f \cap \partial K_g} fg \langle n_{K_g}, Tx \rangle d\mathcal{H}^{n-1} \\ &= \frac{\text{tr } T}{n} \left(\int_{K_f \cap \text{int } K_g} fg \langle \nabla \varphi, x \rangle dx + \int_{K_f \cap \partial K_g} fg \langle n_{K_g}, x \rangle d\mathcal{H}^{n-1} \right), \end{aligned}$$

and the source's centering identity holds:

$$0 = \int_{K_f \cap \text{int } K_g} fg \nabla \varphi dx + \int_{K_f \cap \partial K_g} fg n_{K_g} d\mathcal{H}^{n-1}.$$

3 Proof intuition

The L^2 assumption is stronger than the deformation argument needs. The only bulk vector fields are bounded on compact supports and grow at most linearly on unbounded supports. Every integrable log-concave function has a distributional gradient with an automatically finite first moment. Indeed, its convex superlevel sets grow at most linearly in the logarithmic level; coarea therefore gives an exponentially weighted integral of at most polynomially growing perimeters.

In the compact part of the source proof, the L^2 norm appears only in a shrinking boundary-layer error. Absolute continuity of the automatic L^1 gradient norm makes that error vanish. For unbounded supports, the same weighted L^1 estimate supplies dominated convergence. Nothing changes in the boundary-measure part of the source argument.

4 Proof

We first isolate the automatic estimate.

Lemma 1 (Automatic weighted BV estimate). *Let $0 \neq g = e^{-\varphi} \in \mathcal{L}^n$, extended by zero off its support. Then*

$$\int_{\mathbb{R}^n} (1 + |x|) d|Dg|(x) < \infty. \quad (8)$$

Consequently,

$$\int_{\text{int } K_g} (1 + |x|) |\nabla g(x)| dx < \infty. \quad (9)$$

Moreover, $\int |x|^q g(x) dx < \infty$ for every $q \geq 0$.

Proof. Let $m = \min \varphi$, choose x_0 with $\varphi(x_0) = m$, and put

$$C_s = \{x : \varphi(x) \leq m + s\} \quad (s > 0).$$

The coercivity in the definition of $\mathcal{L}^n$ makes C_1 a convex body. Convexity gives

$$C_s - x_0 \subset \max\{1, s\}(C_1 - x_0). \quad (10)$$

For $s \leq 1$ this is nesting. For $s \geq 1$, if $x \in C_s$, then $\varphi(x_0 + (x - x_0)/s) \leq (1 - s^{-1})m + s^{-1}\varphi(x) \leq m + 1$.

Let $P(C) = \mathcal{H}^{n-1}(\partial C)$ and $R = \max_{C_1} |x - x_0|$. Perimeter is monotone under inclusion for convex bodies [3]; hence (10) yields

$$P(C_s) \leq P(C_1) \max\{1, s\}^{n-1},$$

$$\int_{\partial C_s} (1 + |x|) d\mathcal{H}^{n-1}(x) \leq P(C_1) \max\{1, s\}^{n-1} (1 + |x_0| + R \max\{1, s\}). \quad (11)$$

The coarea characterization of BV functions [2], applied to the superlevel sets of g , and the change of variables $t = e^{-m}e^{-s}$ give

$$\int (1 + |x|) d|Dg|(x) = e^{-m} \int_0^\infty e^{-s} \int_{\partial C_s} (1 + |x|) d\mathcal{H}^{n-1}(x) ds < \infty$$

by (11). This proves (8), and (9) follows from the absolutely continuous part of Dg . Finally, (10) also gives an exponential pointwise tail outside a fixed ball, so all polynomial moments of g are finite. $\square$

We now revisit exactly where (SM2) enters the source proof.

Compact supports. In the proof of the compact variational formula [1, Theorem 17 and Proposition 19], the L^2 hypothesis is used only to bound the term denoted $I_3(j, \varepsilon)$. With the source notation,

$$|I_3(j, \varepsilon)| \leq 2\|f\|_\infty \sup_{\substack{0 \leq \varepsilon \leq \varepsilon_0 \\ x \in K_g}} |A_\varepsilon x| \int_{M_j(g)} |\nabla g(x)| dx. \quad (12)$$

Here the matrices A_ε stay bounded, K_g is compact, and the boundary layers $M_j(g)$ decrease to ∂K_g , which has Lebesgue measure zero. Lemma 1 and absolute continuity of the integral make the right side of (12) tend to zero, uniformly for small ε . All other estimates in that compact proof are independent of (SM2). The same replacement proves the compact translation formula, with $A_\varepsilon x$ replaced by a fixed translation vector.

Unbounded supports. Use the source truncations

$$f_\delta = f e^{-\delta|x|^2} \mathbf{1}_{\{f > \delta\}}, \quad g_\delta = g e^{-\delta|x|^2} \mathbf{1}_{\{g > \delta\}}.$$

They have compact supports, so the already proved L^1 version of the compact formula applies; no L^2 estimate on g_δ is required.

It remains to pass to the limit. For $0 \leq \varepsilon \leq \varepsilon_0$, $\|A_\varepsilon\|$ and $\|T_\varepsilon\|$ are uniformly bounded. Thus the first bulk error over the source's decreasing set $\Delta_\delta(\varepsilon)$ satisfies

$$\left| \int_{\Delta_\delta(\varepsilon)} f(T_\varepsilon x) \langle \nabla g(x), A_\varepsilon x \rangle dx \right| \leq C\|f\|_\infty \int_{\Delta_\delta(\varepsilon)} |x| |\nabla g(x)| dx \longrightarrow 0. \quad (13)$$

The remaining coefficient-difference term is dominated by $C\|f\|_\infty |x| |\nabla g(x)|$ and tends to zero by dominated convergence. Finally, differentiating the Gaussian factor produces a term bounded by

$$C\delta\|f\|_\infty \int_{\mathbb{R}^n} |x|^2 g(x) dx \longrightarrow 0. \quad (14)$$

In fact the convergence in (13) is uniform in ε . Up to null level sets, its lost domain is contained in the union of $\{0 < g \leq \delta\}$ and $T_\varepsilon^{-1}\{0 < f \leq \delta\}$. On the first set use $f \leq \|f\|_\infty$; on the second use $f(T_\varepsilon x) \leq \delta$. Consequently its supremum over the compact parameter interval is at most

$$C\|f\|_\infty \int_{\{0 < g \leq \delta\}} |x| |\nabla g(x)| dx + C\delta \int_{K_g} |x| |\nabla g(x)| dx, \quad (15)$$

which tends to zero by Lemma 1. Uniformity for the coefficient-difference term follows from the same integrable majorant: first discard a tail on which $C\|f\|_\infty|x||\nabla g|$ has small integral, and then use uniform convergence of the Gaussian coefficients on the remaining ball, since the families $T_\varepsilon, A_\varepsilon$ are bounded. The Gaussian bound (14) is already uniform.

The source's estimates for the boundary terms involve ε only through these bounded matrix families and the traces $f(T_\varepsilon \cdot)$. Their compactness/dominated-convergence argument, with boundary continuity supplied by (R1)–(R2), is uniform on the same parameter interval and contains no use of (SM2). Hence that part of the passage to the limit is unchanged. The translation formula is easier: its bulk field is constant and only $\int |\nabla g|$ is needed.

We have therefore proved both first-variation formulas used by Hoehner–Roysdon without (SM2). Their remaining argument differentiates at the maximal position, sets the derivatives equal to zero, and performs linear algebra; it contains no further use of (SM2). This proves Theorem 1.

5 Strictness of the improvement

For $0 < \alpha \leq \frac{1}{2}$, define on $\mathbb{R}$

$$g_\alpha(x) = \begin{cases} x^\alpha, & 0 < x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The extended-valued function $-\log g_\alpha = -\alpha \log x$ on $(0, 1]$ and $+\infty$ elsewhere is convex, so g_α is log-concave. Its interior gradient obeys

$$\int_0^1 |g'_\alpha(x)| dx = 1, \quad \int_0^1 |g'_\alpha(x)|^2 dx = \alpha^2 \int_0^1 x^{2\alpha-2} dx = \infty.$$

Thus the removed condition is genuinely stronger than the automatic estimate.

6 Verifier report

The proof was checked along the following dependency chain.

1. The logarithmic sublevel parameter is normalized so that coarea carries the factor e^{-s} ; the perimeter/radius growth is at most degree n , hence integrable.
2. The source proof was searched for every occurrence of $\int |\nabla g|^2$. In the compact argument it enters only through the I_3 Cauchy–Schwarz estimate, replaced by (12).
3. The compact boundary layer decreases to a convex boundary of Lebesgue measure zero, so L^1 absolute continuity applies.
4. In the unbounded argument, (13) retains the necessary factor $|x|$ and uses the weighted estimate, not an unweighted shortcut.
5. Estimate (14) retains the decaying factor g ; replacing it by $\|g\|_\infty$ would not give a valid limit on all of $\mathbb{R}^n$.
6. Uniformity in the deformation parameter is explicit in (15). The coefficient error is handled by an integrable-tail/compact-ball argument, and the retained boundary-transversality assumptions supply continuity for the unchanged surface-term limit; pointwise dominated convergence alone is not being used as a uniform-convergence principle.

7. The source’s surface-term convergence and both boundary transversality hypotheses are left untouched. No computational experiment is used as proof.

Recommended human-review focus: confirm the source’s claim that the boundary term convergence is independent of (SM2), and check the identical L^1 substitution in its sketched translation formula.

7 Limitations and novelty status

This result removes only the gradient second-moment clause. It does not remove the source’s boundary-intersection hypotheses or alter its regularity/trace framework. A bounded search on 19 July 2026 used arXiv:2401.01033, the exact title, authors, and the phrases “second moment condition,” “bounded second moment,” and “maximal intersection position.” No later resolution or duplicate of this weighted-BV removal was found. The current arXiv v3, dated 26 May 2025, still states Remark 13. This is a bounded novelty check, not an exhaustive priority claim.

References

- [1] S. Hoehner and M. Roysdon, *On maximal intersection position for logarithmically concave functions and measures*, arXiv:2401.01033v3 (2025).
- [2] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, revised ed., CRC Press, 2015.
- [3] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge University Press, 2014.